spread out (Figure 13.6a). Rather than being independent of distance, the force of gravity would vary *inversely* with separation when objects were close enough together.

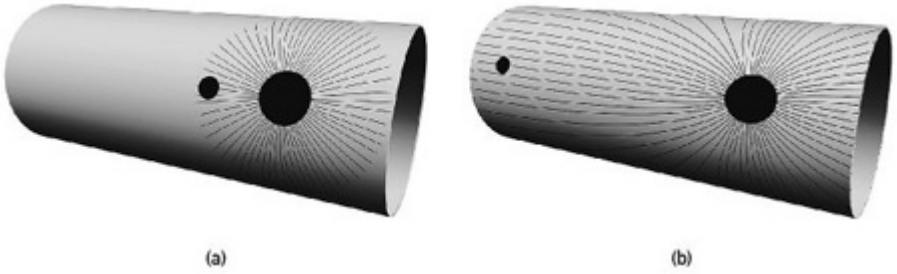


Figure 13.6 **(**a**)** When objects are close, the gravitational pull varies as it does in two space dimensions. **(**b**)** When objects are farther apart, the gravitational pull behaves as it does in one space dimension—it is constant.

Thus, if you were an experimenter in this universe, and you developed exquisitely accurate methods for measuring gravitational attraction, here's what you would find. When two objects were extremely close, much closer than the size of the curled-up dimension, their gravitational attraction would diminish in proportion to their separation, just as you expect for a universe with two space dimensions. But then, when the objects were about as far apart as the circumference of the curled-up dimension, things would change. Beyond this distance, the gravitational field lines would be unable to spread any further. They would have spilled out as far as they could into the second curled-up dimension—they would have saturated that dimension—and so from this distance onward the gravitational force would no longer diminish, as illustrated in Figure 13.6b. You can compare this saturation with the plumbing in an old house. If someone opens the faucet in the kitchen sink when you're just about to rinse the shampoo out of your hair, the water pressure can drop because the water spreads between the two outlets. The pressure will diminish yet again should someone open the faucet in the laundry room, since the water will spread even more. But once all the faucets in the house are open, the pressure will remain constant. Although it might not provide the relaxing, high-water-pressure experience you'd anticipated, the pressure in the shower will not drop any further once the water has completely spread throughout all "extra" outlets. Similarly,